Waseem Raza, PhD (Electrical and Computer Engineering)

ECE Educator-Scholar | Intelligent Sensing, Signal Processing & Wireless AI

Undergraduate Teaching & Mentoring | Hands-On ECE Labs | Applied AI Research

✉ waseem.placements@gmail.com | waseemraza844@gmail.com

New Jersey, United States ☎ +1 732-803-9703

LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio

U.S. work authorization: STEM OPT through Feb 2028 | EB-2 NIW I-140 approved

Academic Profile

PhD in Electrical and Computer Engineering (University of Oklahoma, 2025) with experience teaching undergraduate engineering, supervising labs, mentoring student projects, and conducting applied research. Scholarship connects wireless AI, reproducible experimental platforms, and wearable sensing for health monitoring. Interested in project-based ECE teaching, interdisciplinary sensing systems, and research that actively involves undergraduate students.

Research Interests

Biomedical and Wearable Sensing; Signal Processing and Intelligent Systems; Wireless AI and Networked Cyber-Physical Systems; Robust, Data-Efficient Machine Learning; Experimental Testbeds and Undergraduate Research

Doctoral Research Focus

PhD dissertation focused on AI/ML for wireless networks, including data-efficient radio propagation modeling, generative augmentation, robust learning under distribution shift, network self-healing, and reproducible simulation and experimental evaluation.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*

University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*

University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*

University of Engineering and Technology, Taxila, Pakistan

Relevant Graduate Coursework

LTE Networks and Telecommunications; Emerging 5G Systems; Advanced ML and Computer Vision; Wireless Testbed Design; Numerical Analysis; Information Theory; Digital Signal Processing

Teaching, Research & Professional Experience

Teaching & Instruction

Lecturer, Department of Technology **12/2017 - 09/2019**

University of Lahore, Pakistan *Lahore, Pakistan*

- Taught programming, communication systems, and wireless communication with practical examples.
- Advised senior-design students on projects, academic plans, and early careers.
- Aligned course materials with learning outcomes and accreditation standards.
- Supported accreditation, exams, technical workshops, and student activities.

Teaching areas: Computer Programming; Communication Systems; Signals and Digital Signal Processing; Wireless Communications; Applied Machine Learning; Hands-On Labs and Engineering Design; Sensing and Experimental Data Analysis

Student mentoring and research development

- Mentored undergraduate and graduate students in wireless ML experiments and research writing.
- Guided thesis and capstone projects in modeling, experimental design, and presentation.
- Created labs and assignments connecting communications theory with engineering practice.

University Research Appointments

Graduate Research Assistant **09/2019 - 07/2025**

AI4Networks, University of Oklahoma *Tulsa, OK*

- Built 3GPP-compliant Python simulation tools and an AI-enabled 5G research testbed.
- Built GAN, transfer-learning, autoencoder, and reinforcement-learning models for wireless data.
- Led 100+ experiments testing wireless ML under domain shift and limited labeled data.
- Developed self-healing and outage-management methods, improving resilience by 30%.
- Published wireless AI research and mentored junior researchers in experimental design.

Research Associate, ComSens Lab **06/2016 - 12/2017**
COMSATS University Islamabad, Pakistan *Islamabad, Pakistan*

- Modeled wireless communication systems through simulation, analysis, and technical reports.
- Mentored Bachelor's and Master's theses in experiment design and research presentation.
- Reviewed student reports and manuscripts for rigor and publication readiness.
- Supported proposals, literature reviews, lab seminars, and research coordination.

Full-Time Research Scholar, Communication Systems Lab **08/2014 - 02/2016**
University of Engineering and Technology, Taxila, Pakistan *Taxila, Pakistan*

- Researched wireless communication systems while completing an MSc in Telecommunications.
- Supported communications courses, labs, assignments, grading, and examinations.
- Led wireless communication labs and prepared technical manuals and experiments.
- Developed course and student-project materials for practical wireless engineering.

Industry Research & Engineering

Machine Learning Engineer **02/2025 - Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Applied domain-adaptive ML to telecom service assurance and model validation.
- Designed cross-domain evaluation and error-analysis methods for operational ML.
- Worked with engineering and public-safety teams on reliable network assurance.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**
Impac Exploration Services *Weatherford, OK*

- Built LLM-enabled analytics to interpret geochemical well data for decision support.
- Built Python pipelines in Azure Databricks, Functions, and Synapse Analytics.
- Automated exploration-data ETL, cutting preparation time by 40%.
- Used data-validation and quality controls to improve output reliability by 20%.
- Turned geologists' technical needs into structured datasets and useful analytics.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR logs with ML, reducing diagnostic time by 20%.
- Improved 5G NR-NTN cell selection by 15% through data-driven validation.
- Presented operational data insights to next-generation wireless engineering teams.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved wireless CSI compression by 25% with transfer learning and autoencoders.
- Ran 50+ experiments on wireless ML robustness under changing data conditions.
- Translated research results into recommendations for future wireless networks.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Improved 5G standalone user-timing precision by 20% with ML and statistics.
- Analyzed user traces with SQL and presented results to engineering leadership.
- Turned operational traces into indicators for network diagnosis and optimization.

Research Contributions

Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *HealthINF | Accepted* 2026

AI-Powered Resilience: A Dual-Approach for Outage Management in Dense Cellular Networks *Computer Communications* 2025 | DOI

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *Sensors 2025* 2025 | DOI

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *IEEE PIMRC* 2024 | DOI

Refining Wireless Propagation Models using Domain-Informed GANs amid Data Scarcity *IEEE VTC-Fall 2024* 2024 | Link

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *IEEE SmartNets* 2024

An AI-Enabled Framework to Defend Ingenious MDT-Based Attacks on the Emerging Zero Touch Cellular Networks *IEEE Network, 38(1), 228-237* 2024 | DOI

Positioning Error Impact Compensation through Data-Driven Optimization in User-Centric Networks *IEEE GLOBECOM 2023* 2023 | Link

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) *IEEE Network* 2022 | Link

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *IEEE WCNC* 2022

Unification of RF Energy Harvesting Schemes under Mixed Rayleigh-Rician Fading Channels *AEÜ - International Journal of Electronics and Communications* 2020 | DOI

Research Projects & Platforms

Wearable Intelligence for Health Monitoring *Research Project | 2024-2026*

Developed deep-learning workflows for recognizing medication-taking and dietary activities from smartwatch accelerometer and gyroscope signals in free-living settings. Evaluated windowing, feature extraction, CNN/LSTM-style sequence models, cross-user generalization, and recall-oriented performance for health-monitoring applications.

Tools & Methods: Wearable AI, Smartwatch IMU, Time Series, Deep Learning, CNN/LSTM, Feature Engineering, Cross-User Evaluation, Activity Recognition, Healthcare AI

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

Tools & Methods: Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

TurboRAN Experimental Platform *Research Infrastructure | 2020-2024*

Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.

Tools & Methods: 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

Propagation Models & Digital Twins *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

AI for Telecom Network & Security Optimization *Certification Project | 2026*

Applied AI concepts to telecommunications network optimization, security analysis, and customer-experience improvement through a structured specialization project. Framed telecom use cases around operational data, AI-assisted analysis, optimization opportunities, and practical service-improvement recommendations.

Tools & Methods: AI for Telecommunications, Network Optimization, Telecom Security, Customer Experience, Python, TensorFlow, Operational Analytics, Service Improvement

Research Methods & Technical Expertise

Teaching & Undergraduate Research: Undergraduate engineering instruction, laboratory facilitation, project and thesis mentoring, applied programming, hands-on experimentation, curriculum alignment, and student-centered technical feedback

Sensing, Signals & Intelligent Systems: Smartwatch inertial sensing, time-series analysis, activity recognition, digital signal processing, sensor data collection, wearable health monitoring, cyber-physical data analysis

Experimental ECE & Communications: 3GPP-informed wireless simulation, cellular communication systems, 5G testbed design, network measurements, model-based evaluation, experimental design, and digital twins

Machine Learning & Control-Oriented AI: Deep learning, generative augmentation, transfer and domain-adaptive learning, reinforcement learning, network self-healing and adaptive optimization

Software & Research Platforms: Python, C/C++, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, SQL, Linux, and Git

Academic Service & Recognition

Academic Service

- Developed course materials and supported engineering-program curriculum alignment and accreditation activities.
- Mentored student project teams, contributed to research seminars, and reviewed technical reports and manuscripts.
- Supported departmental workshops, student orientations, and laboratory learning activities.

Honors & Awards

William H. Barkow Scholarship *2024*
University of Oklahoma, Electrical and Computer Engineering

Clyde L. Farrar Endowment Scholarship *2024*
University of Oklahoma

P.H. Robinson Fellowship *2023*
University of Oklahoma

FAS-Charles Forgan Scholarship *2023*
University of Oklahoma

Nettie Vincent Boggs Graduate Endowment *2022*
University of Oklahoma

SHS-Paul D. Massad Scholarship *2022*
University of Oklahoma

Self-Learning

1. Machine Learning Stanford/Coursera | Completed | Mar 2020
Supervised and unsupervised learning, regularization, model evaluation, and practical predictive modeling.

2. Introduction to Deep Learning & Neural Networks with Keras IBM/Coursera | Completed | Aug 2026
Neural networks, training workflows, optimization, loss functions, and model evaluation with Keras.

3. Introduction to Computer Vision with TensorFlow Google/Coursera | Completed | Aug 2024
Image processing and convolutional neural networks for computer vision using TensorFlow.

4. Python for Data Science, AI & Development IBM/Coursera | Completed | Jul 2026
Python programming, data analysis, APIs, and reproducible data-science workflows.

5. 4G Network Fundamentals Mines-Télécom/Coursera | Completed | Aug 2026
LTE radio and core networks, cellular architecture, mobility management, and 4G fundamentals.

6. 5G Network Fundamentals Mines-Télécom/Coursera | In progress (~60%)
5G NR radio, network architecture, service capabilities, and next-generation cellular fundamentals.